

Setting Up Model Data: Predicting density and assessing factors driving marbled murrelet distribution in Puget Sound using hierarchical distance sampling

Code by S.J. Gillman

Required Packages

```
library(ggplot2)
library(sf)
library(tidyverse)
```

Required Files Previously Made or Provided

- track_obs_info.RData
 - Located in DataFiles and contains ydat_tidal and DetectData
 - ydat_tidal has a row for each unique PSU-Segment survey event (8,095).
 - * sID unique numerical value for each row
 - * trip_date unique stratum-psu-segment-date ID
 - * date date of survey
 - * OBSP observer-pair combination for that trip_date
 - * obsNum numerical value for OBSP
 - * month three letter abbreviation for month of survey
 - * dist_shore distance from shore during survey (in meters)
 - * season **b** for “breeding” and **nb** for “non-breeding” seasons.
 - * seasNum numerical value for **season** 1 for **b** or 2 for **nb**
 - * myNum numerical value for the month-year of survey 1 to 100
 - * BF Beaufort Sea State: 0, 1, or 2
 - * totobs total number of MAMU observations
 - * effort total survey area effort (length of trackline times 430 m) in km^2
 - * mNum numerical version of **month** 1 to 11
 - * BF0 1 if BF = 0 otherwise 0
 - * BF1 1 if BF = 1 otherwise 0
 - * BF2 1 if BF = 2 otherwise 0
 - * survey_tide tidal height during survey period (meters)
 - DetectData has observation-level information.
 - * sID unique numerical value for each row
 - * trip_date unique stratum-psu-segment-date ID
 - * date date of survey
 - * obs 1 for every row
 - * grpsz size of group observed
 - * perp_dist perpendicular distance to observation in meters.
 - * OBSP observer-pair combination for that trip_date
 - * obsNum numerical value for OBSP
 - * month three letter abbreviation for month of survey
 - * dist_shore distance from shore during survey (in meters)
 - * season **b** for “breeding” and **nb** for “non-breeding” seasons.
 - * seasNum numerical value for **season** 1 for **b** or 2 for **nb**
 - * myNum numerical value for the month-year of survey 1 to 100
 - * BF Beaufort Sea State: 0, 1, or 2
 - * mNum numerical version of **month** 1 to 11

- * BF0 1 if BF = 0 otherwise 0
 - * BF1 1 if BF = 1 otherwise 0
 - * BF2 1 if BF = 2 otherwise 0
- dynamic_data_clean_grids.rds
 - dynamic_data_clean_tracks.rds
 - all_static_hex.csv
 - all_static_tracks.csv
 - grid_currents.csv
 - track_currents.csv

```
file_path <- "../CovariateProcessing/OUTPUTS/"
## grid observations
load("DataFiles/track_obs_info.RData")
## dynamic covs
GRID_Covs <- readRDS(paste0(file_path,"dynamic_data_clean_grids.rds"))
TRACK_Covs <- readRDS(paste0(file_path,"dynamic_data_clean_tracks.rds"))
## static covs
hex_static_all <- read.csv(paste0(file_path,"all_static_hex.csv"))
track_static_all <- read.csv(paste0(file_path,"all_static_track.csv"))
## Average Current
grid_currents <- read.csv(paste0(file_path,"grid_currents.csv"))
track_currents <- read.csv(paste0(file_path,"track_currents.csv"))
```

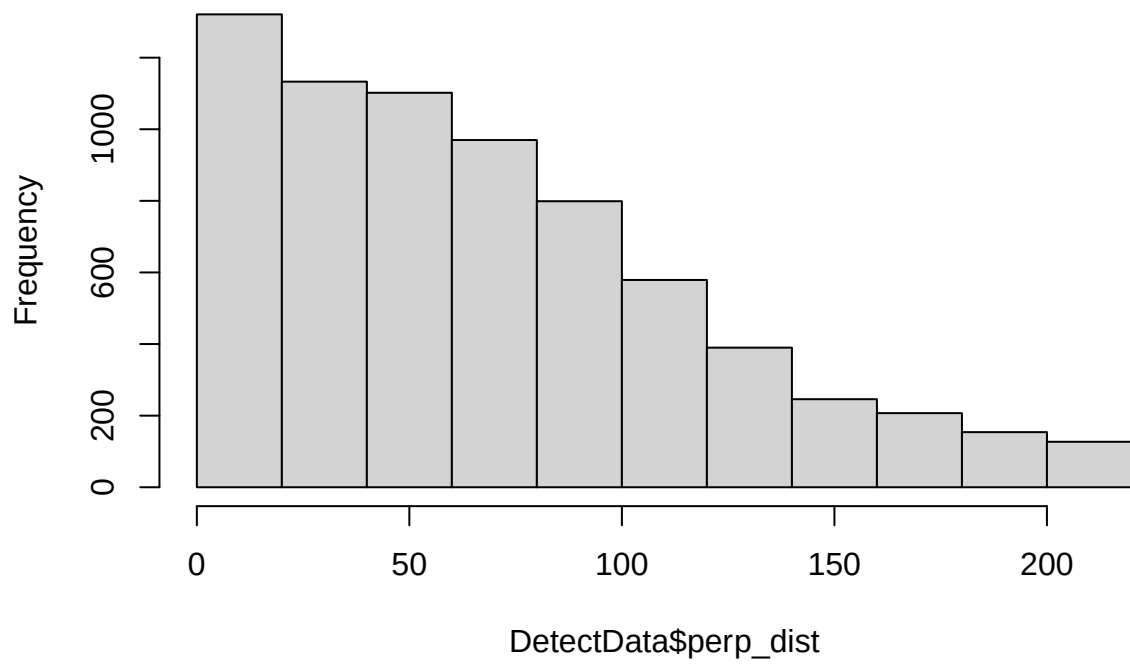
Format Detection Data

```
summary(DetectData$perp_dist, na.rm=T)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      0.00   29.00   59.00   68.52   99.00  215.00
```

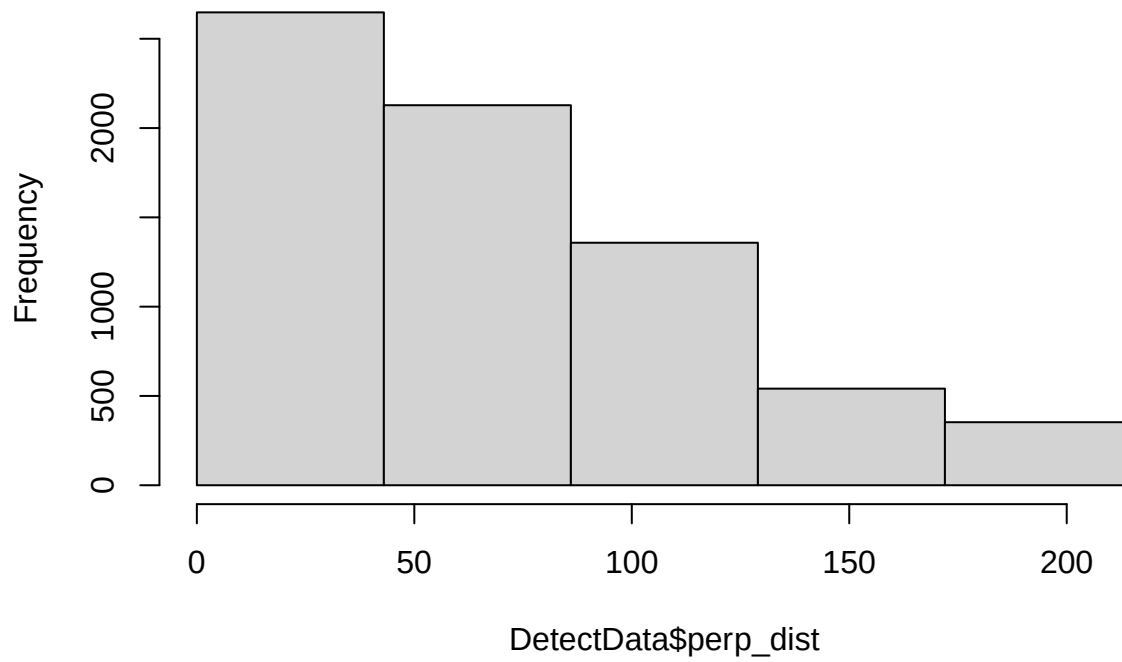
```
hist(DetectData$perp_dist)
```

Histogram of DetectData\$perp_dist

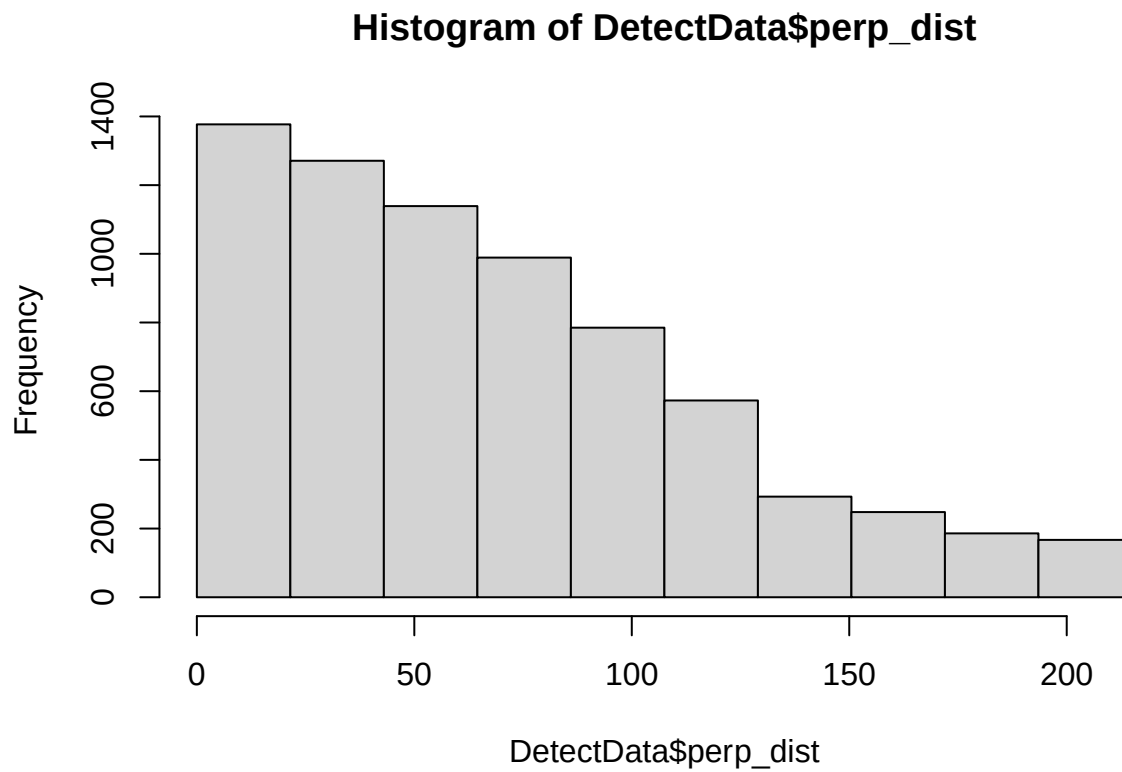


```
custom_breaks <- c(0, 43, 86, 129, 172, 215)
hist(DetectData$perp_dist, breaks = custom_breaks)
```

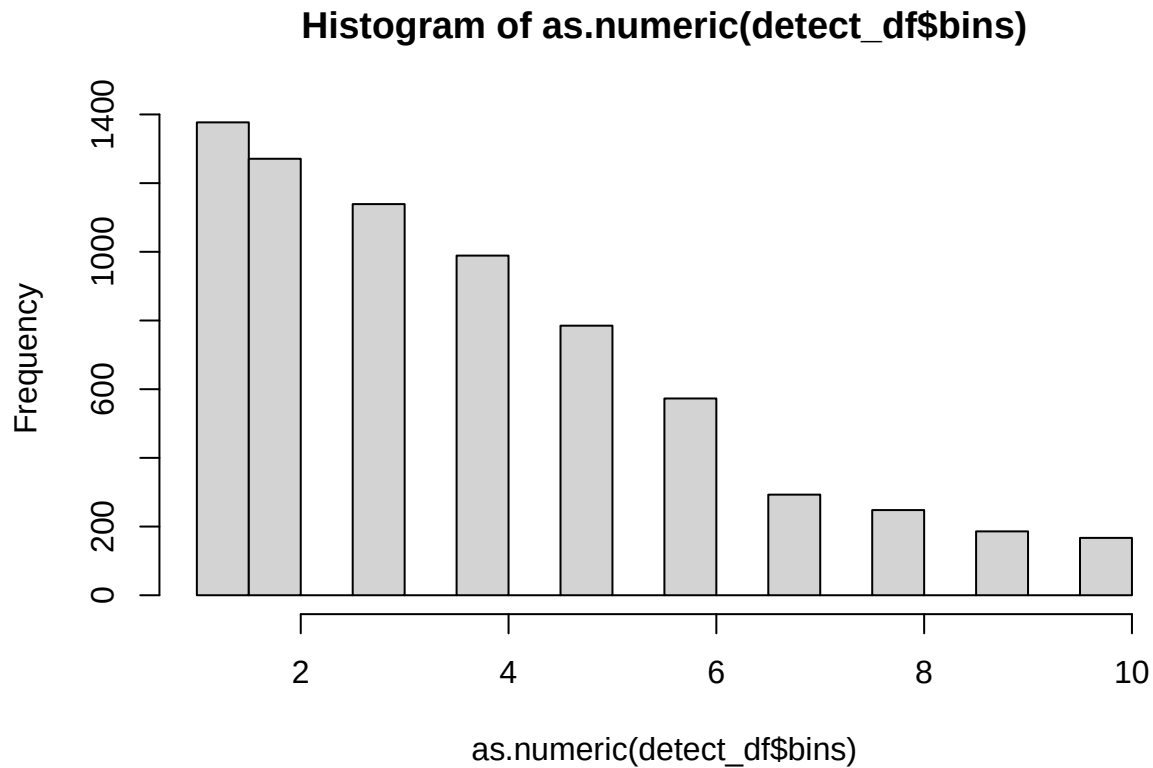
Histogram of DetectData\$perp_dist



```
custom_breaks <- seq(0, 215, by = 21.5)
hist(DetectData$perp_dist, breaks = custom_breaks)
```



```
detect_df <- DetectData %>%  
  rowwise() %>%  
  mutate(bins = as.numeric(cut(perp_dist,  
                                breaks = custom_breaks,  
                                labels = 1:10,  
                                include.lowest = TRUE))) %>%  
  arrange(date,sID)  
  
hist(as.numeric(detect_df$bins))
```



Collinearity Diagnostics

```
diag_col_covs <- hex_static_all %>%
  full_join(grid_currents %>%
    st_drop_geometry(),
    by = c("ID", "numid")) %>%
  st_drop_geometry() %>%
  full_join(GRID_Covs, by = c("ID", "numid")) %>%
  dplyr::select_if(is.numeric) %>%
  dplyr::select(-c(numid, prop_suitable, prop_land, area))

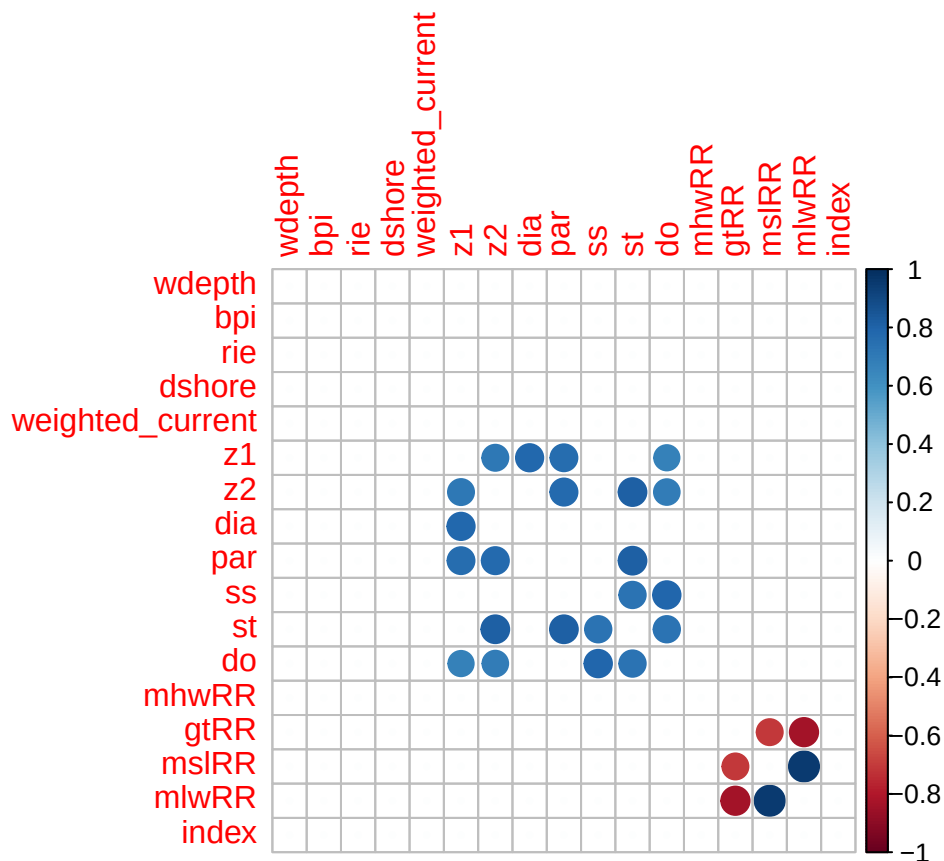
cor_matrix <- round(cor(diag_col_covs,
  use = "pairwise.complete.obs"), 2)
head(cor_matrix)
```

```
##          wdepth  bpi  rie dshore weighted_current  z1  z2  dia
## wdepth          1.00 0.41 -0.35 -0.54          0.25 -0.37 -0.48 -0.27
## bpi              0.41 1.00 0.15 -0.40          -0.10 -0.12 -0.16 -0.09
## rie             -0.35 0.15 1.00 0.06          -0.28 0.21 0.27 0.16
## dshore          -0.54 -0.40 0.06 1.00           0.00 0.20 0.27 0.14
## weighted_current 0.25 -0.10 -0.28 0.00           1.00 -0.12 -0.11 -0.05
## z1              -0.37 -0.12 0.21 0.20          -0.12 1.00 0.71 0.78
##          par    ss    st    do mhwRR  gtRR  ms1RR  mlwRR index
```

```
## wdepth      -0.33 -0.57 -0.53 -0.62 -0.02 -0.14  0.13  0.15  0.09
## bpi         -0.13 -0.24 -0.19 -0.23  0.01  0.10 -0.10 -0.11  0.05
## rie         0.17  0.31  0.29  0.33  0.03  0.20 -0.18 -0.21 -0.01
## dshore      0.24  0.44  0.33  0.41 -0.03 -0.20  0.19  0.21 -0.24
## weighted_current 0.01  0.16 -0.06 -0.08 -0.08 -0.58  0.53  0.61 -0.07
## z1          0.76  0.43  0.63  0.67 -0.34 -0.05 -0.26 -0.19  0.01
```

```
filtered_corr <- ifelse(cor_matrix <= -0.65 &
  cor_matrix !=1 |
  cor_matrix >= 0.65 &
  cor_matrix !=1, cor_matrix, 0)

corrplot::corrplot(filtered_corr,insig="blank")
```



```
filtered_corr %>%
  as.data.frame() %>%
  filter(rowSums(across(where(is.numeric)))!=0) %>%
  select_if(~ !is.numeric(.) || sum(.) != 0)
```

```
##      z1  z2  dia  par  ss  st  do  gtRR  mslRR  mlwRR
## z1    0.00 0.71 0.78 0.76 0.00 0.00 0.67 0.00 0.00 0.00
## z2    0.71 0.00 0.00 0.77 0.00 0.81 0.69 0.00 0.00 0.00
## dia    0.78 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00
## par    0.76 0.77 0.00 0.00 0.00 0.81 0.00 0.00 0.00 0.00
## ss     0.00 0.00 0.00 0.00 0.00 0.73 0.79 0.00 0.00 0.00
```

```
## st      0.00 0.81 0.00 0.81 0.73 0.00 0.73 0.00 0.00 0.00
## do      0.67 0.69 0.00 0.00 0.79 0.73 0.00 0.00 0.00 0.00
## gtRR    0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 -0.71 -0.84
## mslRR   0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 -0.71 0.00 0.95
## mlwRR   0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 -0.84 0.95 0.00
```

Correlations - Z1 - Z2 - PAR - DIA - DO

Wont use Z1

- **Z2**
- ~~Z1~~
- PAR
- ST
- DO

Wont use Z2

- **DIA**
- ~~Z1~~

Even though it doesn't correlate with anything else, reducing number of covs so wont use DIA

- **PAR**
- ~~Z1~~
- ~~Z2~~
- ~~ST~~ Wont use ST because PAR is for diet rep
- **Sea Salinity**
- ~~ST~~
- DO

Wont use Sea Sal because want DO

- **Sea Temp**
- ~~Z2~~
- PAR
- ~~SS~~
- DO

Wont use Temp

- **GT**
- MSL
- MLW

Will use grand tidal range and not MSL or MLW

```
diag_col_covs <- track_static_all %>%
  full_join(track_currents %>%
    st_drop_geometry(),
    by = c("trip_date", "date")) %>%
  st_drop_geometry() %>%
  full_join(TRACK_Covs, by = c("trip_date", "date")) %>%
  full_join(ydat_tidal %>%
    select(c(trip_date, date, dist_shore))) %>%
  dplyr::select_if(is.numeric) %>%
  dplyr::select(-c(prop_suitable, prop_land, total_length, numid))
```

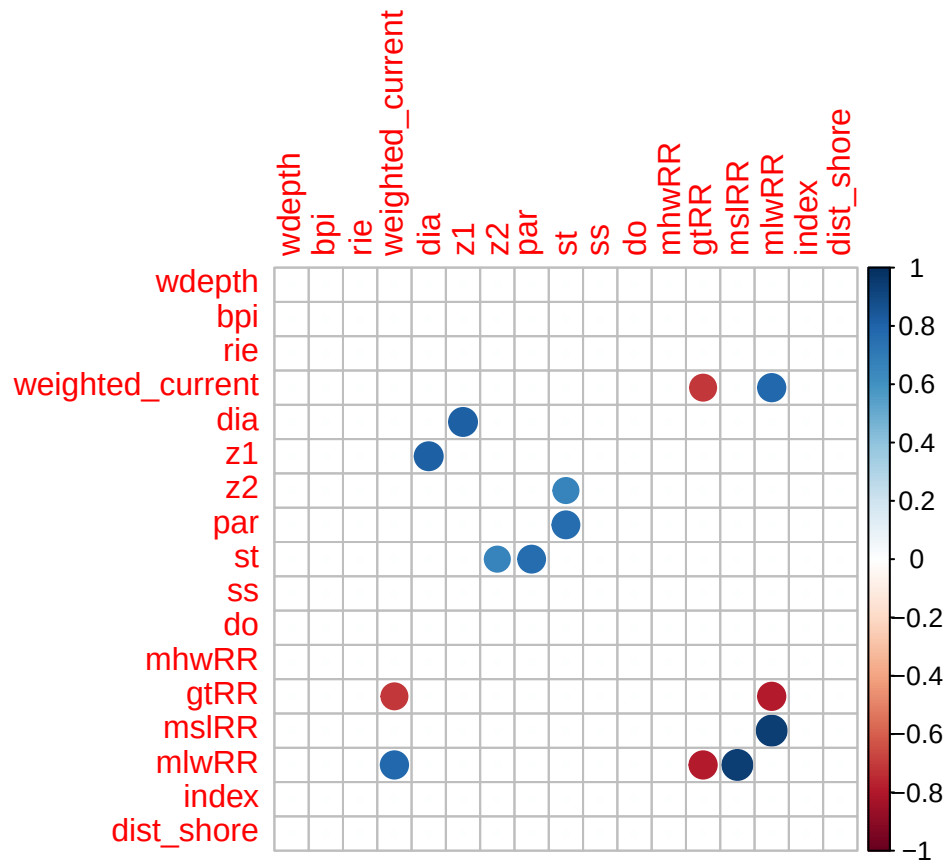
```
## Joining with 'by = join_by(trip_date, date)'
```

```
cor_matrix <- round(cor(diag_col_covs,
  use = "pairwise.complete.obs"), 2)
head(cor_matrix)
```

```
##           wdepth  bpi  rie weighted_current  dia  z1  z2  par
## wdepth           1.00 0.45 -0.34           0.29 -0.10 -0.20 -0.26 -0.10
## bpi              0.45 1.00 0.29           -0.12 -0.09 -0.11 -0.23 -0.10
## rie             -0.34 0.29 1.00           -0.34 0.00 0.00 -0.01 -0.04
## weighted_current 0.29 -0.12 -0.34           1.00 0.01 -0.09 0.01 0.15
## dia             -0.10 -0.09 0.00           0.01 1.00 0.82 0.42 0.36
## z1              -0.20 -0.11 0.00           -0.09 0.82 1.00 0.49 0.64
##           st    ss    do mhwRR  gtRR  mslRR  mlwRR index dist_shore
## wdepth      -0.24 -0.26 -0.49 -0.04 -0.21 0.19 0.23 -0.17      -0.52
## bpi         -0.18 -0.27 -0.32 0.02 0.07 -0.05 -0.06 0.02      -0.50
## rie          0.05 0.12 0.05 0.08 0.22 -0.17 -0.23 0.13       0.12
## weighted_current 0.05 0.33 -0.17 -0.15 -0.71 0.63 0.78 -0.15      -0.12
## dia          0.01 0.12 0.44 -0.23 -0.25 -0.07 -0.01 -0.04       0.07
## z1           0.37 0.12 0.53 -0.35 -0.11 -0.32 -0.24 0.10       0.07
```

```
filtered_corr <- ifelse(cor_matrix <= -0.65 &
  cor_matrix != 1 |
  cor_matrix >= 0.65 &
  cor_matrix != 1, cor_matrix, 0)

corrplot::corrplot(filtered_corr, insig="blank")
```



```
filtered_corr %>%
  as.data.frame() %>%
  filter(rowSums(across(where(is.numeric)))!=0) %>%
  select_if(~ !is.numeric(.) || sum(.) != 0)
```

```
##           weighted_current dia  z1  z2  par  st  gtRR mslRR mlwRR
## weighted_current          0.00 0.00 0.00 0.00 0.00 0.00 -0.71  0.00  0.78
## dia                      0.00 0.00 0.82 0.00 0.00 0.00  0.00  0.00  0.00
## z1                       0.00 0.82 0.00 0.00 0.00 0.00  0.00  0.00  0.00
## z2                       0.00 0.00 0.00 0.00 0.00 0.66  0.00  0.00  0.00
## par                      0.00 0.00 0.00 0.00 0.00 0.76  0.00  0.00  0.00
## st                       0.00 0.00 0.00 0.66 0.76 0.00  0.00  0.00  0.00
## gtRR                     -0.71 0.00 0.00 0.00 0.00 0.00  0.00  0.00 -0.78
## mslRR                    0.00 0.00 0.00 0.00 0.00 0.00  0.00  0.00  0.94
## mlwRR                    0.78 0.00 0.00 0.00 0.00 0.00 -0.78  0.94  0.00
```

Current has a negative correlation with grand tide...

Center & Scale Static Data

```
Static_Scaled <- track_static_all %>%
  full_join(track_currents %>%
    st_drop_geometry(),
```

```

    by = c("trip_date", "date")) %>%
st_drop_geometry() %>%
  select(-total_length) %>%
mutate(wdepth = wdepth*-1) %>%
mutate(depth_scale = scale(wdepth)[,1],
      bpi_scale = scale(bpi)[,1],
      rie_scale = scale(rie)[,1],
      current_scale = scale(weighted_current)[,1]) %>%
mutate(dummy = 1, SHORE = as.numeric(as.factor(shore_code))) %>%
pivot_wider(names_from = shore_code,
            values_from = dummy,
            values_fill = list(dummy = 0)) %>%
mutate(cmec_abb = ifelse(
  new_cmecs == "Coarse Unconsolidated Substrate", "CUS",
  ifelse(new_cmecs == "Fine Unconsolidated Substrate", "FUS",
  ifelse(new_cmecs == "Unconsolidated Mineral Substrate",
    "UMS", gsub(" ", "", new_cmecs)))) %>%
dplyr::select(-new_cmecs) %>%
mutate(dummy = 1, FLOOR = as.numeric(as.factor(cmec_abb))) %>%
pivot_wider(names_from = cmec_abb,
            values_from = dummy,
            values_fill = list(dummy = 0)) %>%
dplyr::select(-c(wdepth, bpi, rie, weighted_current, numid))

```

```
anyNA(Static_Scaled)
```

```
## [1] FALSE
```

```
sapply(Static_Scaled, function(x) sum(is.na(x)))
```

```
##      trip_date      date  depth_scale    bpi_scale    rie_scale
##          0          0          0          0          0
## current_scale    SHORE          MI          OCN EI-Mix_Coarse
##          0          0          0          0          0
##      EI-Gravel    EI-Sand EI-Artificial  EI-Boulder  EI-Mix_Fine
##          0          0          0          0          0
##      FLOOR    MuddySand          Sand          Mud          Gravel
##          0          0          0          0          0
##          CUS    Cobble    SandyMud          UMS          FUS
##          0          0          0          0          0
```

```
table(track_static_all$new_cmecs)
```

```
##
## Coarse Unconsolidated Substrate          Cobble
##                                167          286
## Fine Unconsolidated Substrate          Gravel
##                                402          306
##                                Mud          Muddy Sand
##                                1086         2419
```

```
##                               Sand                               Sandy Mud
##                               2122                               911
## Unconsolidated Mineral Substrate
##                               396
```

```
table(track_static_all$shore_code)
```

```
##
## EI-Artificial    EI-Boulder    EI-Gravel EI-Mix_Coarse    EI-Mix_Fine
##           203           206           382           2619           961
##           EI-Sand           MI           OCN
##           715           314           2695
```

Center & Scale Dynamic Data

All grids in all months and years

```
All_scaled_covs <- TRACK_Covs %>%
  full_join(ydat_tidal,
    by = c("trip_date", "date")) %>%
  ungroup() %>%
  mutate(suit_scale = scale(index)[,1],
    tidal_scale = scale(survey_tide)[,1],
    shore_scale = scale(dist_shore)[,1],
    shoreSQ_scale = shore_scale*shore_scale) %>%
  full_join(Static_Scaled,
    by = c("trip_date", "date")) %>%
  group_by(mon) %>%
  mutate(do_scale = scale(do)[,1],
    par_scale = scale(par)[,1],
    gt_scale = scale(gtRR)[,1]) %>%
  ungroup() %>%
  dplyr::select(c(trip_date, date,
    myNum,
    monyear,
    mNum,
    mon,
    sID,
    # vary by season
    par_scale,
    do_scale,
    gt_scale,
    suit_scale,
    shore_scale,
    shoreSQ_scale,
    # constant
    depth_scale,
    bpi_scale,
    rie_scale,
    # shoreline EI-Mix_Coarse is reference
    `EI-Sand`,
```

```

    `EI-Mix_Fine`,
    `EI-Boulder`,
    `EI-Artificial`,
    `EI-Gravel`,
    MI,
    OCN,
    # substrate Muddy Sand is reference
    Gravel,
    Mud,
    Sand,
    SandyMud,
    UMS,
    FUS,
    CUS,
    Cobble, SHORE, FLOOR)) %>%
ungroup()

sapply(All_scaled_covs, function(x) sum(is.na(x)))

```

```

##      trip_date      date      myNum      monyear      mNum
##      0            0          0          0          0
##      mon          sID      par_scale      do_scale      gt_scale
##      0            0          0          0          342
##      suit_scale    shore_scale shoreSQ_scale depth_scale bpi_scale
##      0            0          0          0          0
##      rie_scale      EI-Sand    EI-Mix_Fine    EI-Boulder EI-Artificial
##      0            0          0          0          0
##      EI-Gravel      MI          OCN          Gravel      Mud
##      0            0          0          0          0
##      Sand          SandyMud      UMS          FUS          CUS
##      0            0          0          0          0
##      Cobble        SHORE        FLOOR
##      0            0          0

```

```

ydat_tidal2 <- ydat_tidal %>%
  ungroup() %>%
  mutate(tidal_scale = scale(survey_tide)[,1]) %>%
  arrange(myNum, sID) %>%
  group_by(myNum) %>%
  mutate(site_my = rank(sID)) %>%
  ungroup() %>%
  full_join(All_scaled_covs %>%
    select(-mon), by = c("sID", "trip_date",
                        "date", "myNum", "mNum")) %>%

  select(-survey_tide) %>%
  arrange(myNum, sID) %>%
  mutate(sID = as.numeric(as.factor(trip_date)),
         obsNum = as.numeric(as.factor(OBSP))) %>%
  arrange(myNum, sID)

detect_df_new <- detect_df %>%
  select(-c(obsNum, sID, myNum)) %>%

```

```

left_join(ydat_tidal2 %>%
  select(c(trip_date,date,sID, obsNum, myNum)),
  by = c("trip_date", "date")) %>%
arrange(myNum, sID)

```

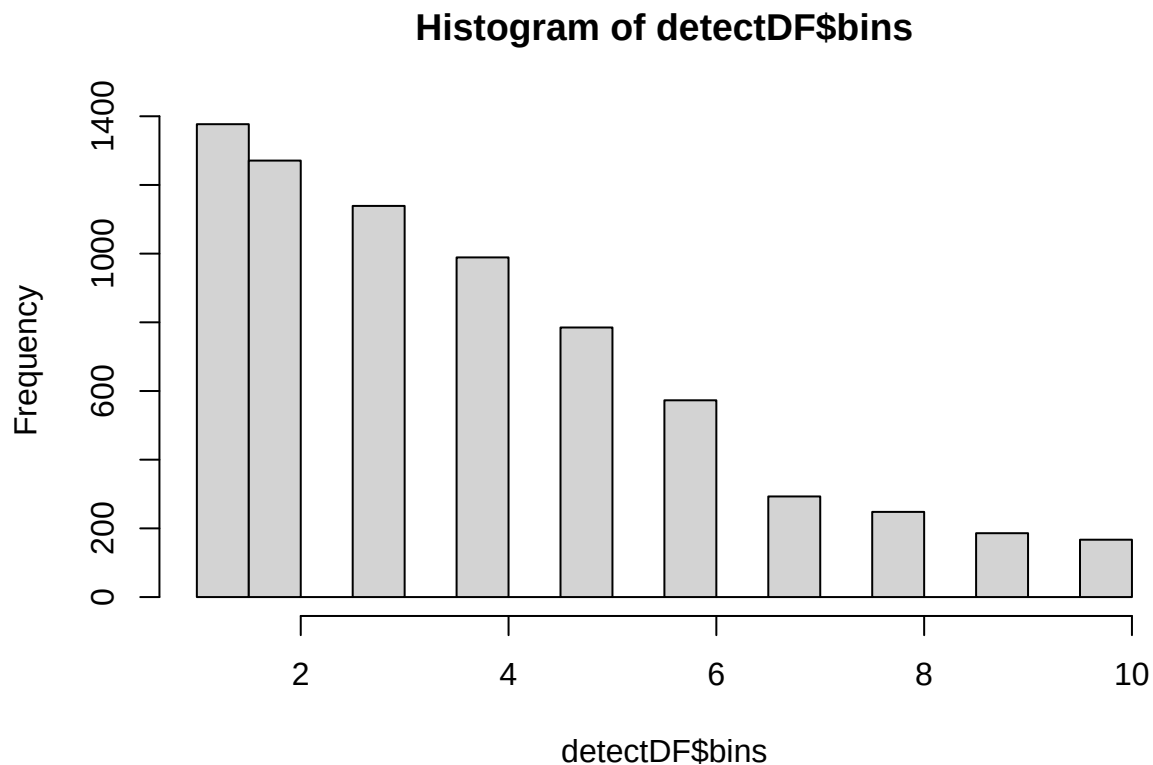
Arrange Data

```

detectDF <- detect_df_new %>%
  arrange(myNum, sID) %>%
  select(c(trip_date,date, sID,grpsz, bins, perp_dist))

hist(detectDF$bins)

```



```

sum_by_survey <- detectDF %>%
  arrange(date) %>%
  mutate(gs_minus1 = grpsz - 1L) %>%
  group_by(sID) %>%
  summarise(sum_gsminus1 = sum(gs_minus1), .groups = "drop") %>%
  ungroup()

ydat <- ydat_tidal2 %>%
  arrange(myNum, sID) %>%

```

```

left_join(sum_by_survey, by = "sID") %>%
mutate(sum_gsminus1 = tidyr::replace_na(sum_gsminus1, 0L),
      yNum = case_when(startsWith(as.character(date), "2012") ~1,
                        startsWith(as.character(date), "2013") ~2,
                        startsWith(as.character(date), "2014") ~3,
                        startsWith(as.character(date), "2015") ~4,
                        startsWith(as.character(date), "2016") ~5,
                        startsWith(as.character(date), "2017") ~6,
                        startsWith(as.character(date), "2018") ~7,
                        startsWith(as.character(date), "2019") ~8,
                        startsWith(as.character(date), "2020") ~9,
                        startsWith(as.character(date), "2021") ~10,
                        startsWith(as.character(date), "2022") ~11,
                        startsWith(as.character(date), "2023") ~12,
                        startsWith(as.character(date), "2024") ~13)) %>%

arrange(myNum, sID) %>%
select(c(trip_date, date, sID,
        totobs, effort, sum_gsminus1,
        obsNum, seasNum, myNum,
        yNum, mNum, BF1,
        BF2, tidal_scale,
        par_scale, do_scale, suit_scale,
        gt_scale, shore_scale, shoreSQ_scale,
        depth_scale, bpi_scale,
        rie_scale,
        `EI-Sand`, `EI-Mix_Fine`, `EI-Boulder`,
        `EI-Artificial`, `EI-Gravel`, `MI`, `OCN`,
        Gravel, Mud, Sand, SandyMud, UMS, FUS, CUS, Cobble, SHORE, FLOOR)) %>%
arrange(myNum, sID)

survey_info <- ydat_tidal2 %>%
mutate(yNum = case_when(startsWith(as.character(date), "2012") ~1,
                        startsWith(as.character(date), "2013") ~2,
                        startsWith(as.character(date), "2014") ~3,
                        startsWith(as.character(date), "2015") ~4,
                        startsWith(as.character(date), "2016") ~5,
                        startsWith(as.character(date), "2017") ~6,
                        startsWith(as.character(date), "2018") ~7,
                        startsWith(as.character(date), "2019") ~8,
                        startsWith(as.character(date), "2020") ~9,
                        startsWith(as.character(date), "2021") ~10,
                        startsWith(as.character(date), "2022") ~11,
                        startsWith(as.character(date), "2023") ~12,
                        startsWith(as.character(date), "2024") ~13),
        year = str_sub(as.character(date), 1, 4)) %>%
select(-c(contains("_scale"), contains("EI-"), MI,
        OCN, Gravel, Mud, Sand, SandyMud, UMS, FUS, CUS, Cobble,
        OBSP, site_my, BF2, BF1)) %>%
left_join(ydat_tidal %>%
        select(survey_tide, trip_date),
        by = "trip_date") %>%
left_join(TRACK_Covs %>%
        select(c(trip_date, date, par, do, gtRR, index)) %>%

```

```

        rename(grand_range = gtRR,
               nest_hab = index),
        by = c("trip_date", "date")) %>%
left_join(track_static_all %>%
          st_drop_geometry() %>%
          mutate(depth = wdepth*-1) %>%
          select(c(trip_date, date, depth, bpi, rie,
                  new_cmecs, shore_code)) %>%
          rename(shoretype = shore_code, seafloor = new_cmecs),
          by = c("trip_date", "date")) %>%
select(c(trip_date, date, sID,totobs, effort,
        month, mNum, monyear, myNum, year, yNum,
        seasNum,obsNum, BF, survey_tide,
        par, do,grand_range,dist_shore, nest_hab,
        depth, bpi,rie, seafloor, shoretype, SHORE, FLOOR))

grid_info <- GRID_Covs %>%
select(c(ID,numid,mon, year,monyear,
        par, do, gtRR, index)) %>%
full_join(hex_static_all %>%
          st_drop_geometry() %>%
          mutate(depth = wdepth*-1) %>%
          rename(shoretype = shore_code,
                 seafloor = new_cmecs,
                 dist_shore = dshore),
          by =c("ID", "numid")) %>%
rename(grand_range = gtRR,
       nest_hab = index) %>%
left_join(survey_info %>%
          distinct(monyear, myNum, mNum, yNum, seasNum),
          by = "monyear") %>%
left_join(survey_info %>%
          distinct(seafloor, FLOOR),
          by = "seafloor") %>%
left_join(survey_info %>%
          distinct(shoretype, SHORE),
          by = "shoretype") %>%
select(c(ID, numid, area,
        mon, mNum, monyear, myNum, year, yNum,
        seasNum,par, do,grand_range,
        dist_shore, nest_hab,
        depth, bpi,rie, seafloor, shoretype, SHORE,FLOOR))

```

Save

```

save(ydat, custom_breaks,detectDF,file ="formatted_data_2026.RData")

save(survey_info, grid_info,file ="Survey_Grid_Info.RData")

```

Summary Information

Track-level information

```
sum(ydat_tidal$effort)
```

```
## [1] 22829.8
```

```
ydat_tidal %>%  
  mutate(year = str_sub(date, 1,4)) %>%  
  group_by(year,season) %>%  
  summarise(tot = round(sum(effort),0), .groups="drop")
```

```
## # A tibble: 21 x 3  
##   year season tot  
##   <chr> <chr> <dbl>  
## 1 2012 b      523  
## 2 2012 nb     772  
## 3 2013 b      496  
## 4 2013 nb    1536  
## 5 2014 b      492  
## 6 2014 nb    1464  
## 7 2015 b      487  
## 8 2015 nb    1974  
## 9 2016 b      489  
## 10 2016 nb    1970  
## # i 11 more rows
```

```
ydat_tidal %>%  
  mutate(year = str_sub(date, 1,4)) %>%  
  group_by(year) %>%  
  summarise(tot = round(sum(effort),0), .groups="drop")
```

```
## # A tibble: 13 x 2  
##   year tot  
##   <chr> <dbl>  
## 1 2012 1295  
## 2 2013 2033  
## 3 2014 1956  
## 4 2015 2461  
## 5 2016 2458  
## 6 2017 1964  
## 7 2018 2321  
## 8 2019 1971  
## 9 2020 2044  
## 10 2021 974  
## 11 2022 1137  
## 12 2023 1547  
## 13 2024 669
```

```
ydat_tidal %>%
  mutate(year = str_sub(date, 1,4)) %>%
  group_by(year,month) %>%
  summarise(tot = sum(totobs), av = round(mean(totobs),2), .groups="drop") %>%
  ungroup() %>%
  filter(month == "JAN")
```

```
## # A tibble: 11 x 4
##   year month tot av
##   <chr> <chr> <dbl> <dbl>
## 1 2013 JAN 145 1.24
## 2 2014 JAN 130 0.96
## 3 2015 JAN 116 0.87
## 4 2016 JAN 219 1.62
## 5 2017 JAN 43 0.42
## 6 2018 JAN 110 0.81
## 7 2019 JAN 108 0.77
## 8 2020 JAN 10 0.1
## 9 2021 JAN 166 1.12
## 10 2023 JAN 255 2.45
## 11 2024 JAN 34 0.52
```

```
ydat_tidal %>%
  mutate(year = str_sub(date, 1,4)) %>%
  group_by(year,month) %>%
  summarise(tot = sum(totobs), av = round(mean(totobs),2), .groups="drop") %>%
  ungroup() %>%
  filter(month == "FEB")
```

```
## # A tibble: 11 x 4
##   year month tot av
##   <chr> <chr> <dbl> <dbl>
## 1 2013 FEB 93 0.98
## 2 2014 FEB 16 0.28
## 3 2015 FEB 42 0.71
## 4 2016 FEB 12 0.16
## 5 2017 FEB 31 0.29
## 6 2018 FEB 25 0.46
## 7 2019 FEB 18 0.23
## 8 2020 FEB 234 2.75
## 9 2021 FEB 46 0.43
## 10 2023 FEB 71 1.18
## 11 2024 FEB 105 1.72
```

```
ydat_tidal %>%
  mutate(year = str_sub(date, 1,4)) %>%
  group_by(year,month) %>%
  summarise(tot = sum(totobs), av = round(mean(totobs),2), .groups="drop") %>%
  ungroup() %>%
  filter(month == "MAR")
```

```
## # A tibble: 9 x 4
```

```
##   year month   tot   av
##   <chr> <chr> <dbl> <dbl>
## 1 2015  MAR    159  1.07
## 2 2016  MAR    135  1.02
## 3 2017  MAR     48  0.31
## 4 2018  MAR    125  0.93
## 5 2019  MAR     34  0.24
## 6 2020  MAR      3  0.06
## 7 2021  MAR    106  0.83
## 8 2023  MAR     26  0.27
## 9 2024  MAR     20  0.21
```

```
ydat_tidal %>%
  mutate(year = str_sub(date, 1,4)) %>%
  group_by(year,month) %>%
  summarise(tot = sum(totobs), av = round(mean(totobs),2), .groups="drop") %>%
  ungroup() %>%
  filter(month == "APR")
```

```
## # A tibble: 5 x 4
##   year month   tot   av
##   <chr> <chr> <dbl> <dbl>
## 1 2015  APR     24  0.59
## 2 2016  APR     17  0.4
## 3 2017  APR      0  0
## 4 2018  APR     20  0.34
## 5 2019  APR      3  0.06
```

```
ydat_tidal %>%
  mutate(year = str_sub(date, 1,4)) %>%
  group_by(year,month) %>%
  summarise(tot = sum(totobs), av = round(mean(totobs),2), .groups="drop") %>%
  ungroup() %>%
  filter(month == "MAY")
```

```
## # A tibble: 8 x 4
##   year month   tot   av
##   <chr> <chr> <dbl> <dbl>
## 1 2012  MAY     69  1.64
## 2 2013  MAY     25  1.47
## 3 2014  MAY     30  0.81
## 4 2015  MAY     46  2.71
## 5 2016  MAY      6  0.18
## 6 2018  MAY     33  2.06
## 7 2020  MAY     17  0.53
## 8 2022  MAY     15  0.47
```

```
ydat_tidal %>%
  mutate(year = str_sub(date, 1,4)) %>%
  group_by(year,month) %>%
  summarise(tot = sum(totobs), av = round(mean(totobs),2), .groups="drop") %>%
  ungroup() %>%
  filter(month == "JUN")
```

```
## # A tibble: 8 x 4
##   year month   tot   av
##   <chr> <chr> <dbl> <dbl>
## 1 2012 JUN     25 0.66
## 2 2013 JUN     87 1.61
## 3 2014 JUN     57 1.68
## 4 2015 JUN    103 2.06
## 5 2016 JUN     42 0.98
## 6 2018 JUN    150 2.14
## 7 2020 JUN     77 2.26
## 8 2022 JUN     82 1.67
```

```
ydat_tidal %>%
  mutate(year = str_sub(date, 1,4)) %>%
  group_by(year,month) %>%
  summarise(tot = sum(totobs), av = round(mean(totobs),2), .groups="drop") %>%
  ungroup() %>%
  filter(month == "JUL")
```

```
## # A tibble: 8 x 4
##   year month   tot   av
##   <chr> <chr> <dbl> <dbl>
## 1 2012 JUL    129 3.22
## 2 2013 JUL     26 0.54
## 3 2014 JUL     60 1.22
## 4 2015 JUL     63 1.19
## 5 2016 JUL    155 3.52
## 6 2018 JUL     54 1.59
## 7 2020 JUL    122 2.26
## 8 2022 JUL     51 1.31
```

```
ydat_tidal %>%
  mutate(year = str_sub(date, 1,4)) %>%
  group_by(year,month) %>%
  summarise(tot = sum(totobs), av = round(mean(totobs),2), .groups="drop") %>%
  ungroup() %>%
  filter(month == "SEP")
```

```
## # A tibble: 7 x 4
##   year month   tot   av
##   <chr> <chr> <dbl> <dbl>
## 1 2014 SEP     24 0.31
## 2 2015 SEP     39 0.49
## 3 2016 SEP      4 0.06
## 4 2017 SEP      4 0.11
## 5 2018 SEP     18 0.72
## 6 2019 SEP      6 0.06
## 7 2020 SEP      4 0.06
```

```
ydat_tidal %>%
  mutate(year = str_sub(date, 1,4)) %>%
  group_by(year,month) %>%
```

```
summarise(tot = sum(totobs), av = round(mean(totobs),2), .groups="drop") %>%
ungroup() %>%
filter(month == "OCT")
```

```
## # A tibble: 11 x 4
##   year month tot av
##   <chr> <chr> <dbl> <dbl>
## 1 2012 OCT    10 0.21
## 2 2013 OCT   112 0.65
## 3 2014 OCT    44 0.4
## 4 2015 OCT    47 0.43
## 5 2016 OCT   144 1.09
## 6 2017 OCT    50 0.32
## 7 2018 OCT    30 0.24
## 8 2019 OCT    36 0.71
## 9 2020 OCT    40 0.33
## 10 2022 OCT    71 0.93
## 11 2023 OCT   138 1.44
```

```
ydat_tidal %>%
mutate(year = str_sub(date, 1,4)) %>%
group_by(year,month) %>%
summarise(tot = sum(totobs), av = round(mean(totobs),2), .groups="drop") %>%
ungroup() %>%
filter(month == "NOV")
```

```
## # A tibble: 11 x 4
##   year month tot av
##   <chr> <chr> <dbl> <dbl>
## 1 2012 NOV   194 1.2
## 2 2013 NOV   234 1.58
## 3 2014 NOV   212 1.89
## 4 2015 NOV   123 0.92
## 5 2016 NOV    83 0.54
## 6 2017 NOV   118 1.03
## 7 2018 NOV    49 0.64
## 8 2019 NOV    73 0.58
## 9 2020 NOV    97 0.79
## 10 2022 NOV    56 0.8
## 11 2023 NOV    85 1
```

```
ydat_tidal %>%
mutate(year = str_sub(date, 1,4)) %>%
group_by(year,month) %>%
summarise(tot = sum(totobs), av = round(mean(totobs),2), .groups="drop") %>%
ungroup() %>%
filter(month == "DEC")
```

```
## # A tibble: 11 x 4
##   year month tot av
##   <chr> <chr> <dbl> <dbl>
## 1 2012 DEC   118 1.28
```

```
## 2 2013 DEC 40 0.65
## 3 2014 DEC 43 0.54
## 4 2015 DEC 56 1.02
## 5 2016 DEC 8 0.25
## 6 2017 DEC 45 0.73
## 7 2018 DEC 62 0.57
## 8 2019 DEC 41 0.59
## 9 2020 DEC 88 1.33
## 10 2022 DEC 64 1.05
## 11 2023 DEC 45 0.8
```

```
sum(ydat_tidal$totobs)
```

```
## [1] 7028
```